

# Ahmed Fathy

Software Engineer

ahmedfathi20044002@gmail.com | +20 155 285 1443 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Haram, Giza, Egypt 12556

## EDUCATION

- **Cairo University, Faculty of Engineering** Giza, Egypt  
*B.Sc. in Computer Engineering; GPA: 3.7* 2022 – 2027

## EXPERIENCE

- **Al Nada Scientific Office** | *Software Developer (Part-time)* Cairo, Egypt | June 2022 – 2025
  - **ERP System:** Developed a local desktop application (packaged with Electron) managing customers, suppliers, inventory, and sales records using React.js, Express.js, and PostgreSQL.
  - **Corporate Web Solution:** Built the official company website integrated with an e-commerce platform designed to seamlessly manage and showcase sponsors, partners, services, and dynamic product catalogs.

## PROJECTS

- **5-Stage Pipelined Processor (RISC)** | *VHDL, QuestaSim, Computer Architecture* 2025
  - **Core Architecture:** Engineered a 32-bit 5-stage pipelined processor in VHDL, supporting a custom ISA of over 25 instructions.
  - **Hazard Resolution:** Optimized throughput by implementing Hazard Detection and Data Forwarding units to resolve control and data dependencies.
  - **Verification:** Validated functional correctness through rigorous simulation and testbench environments in QuestaSim.
- **CNN Convolution Accelerator** | *Verilog, OpenLane2, ASIC Design* 2025
  - **Architecture:** Designed a synthesizable 2D convolution accelerator using Systolic Arrays and fully pipelined dataflow to maximize MAC throughput.
  - **Memory Hierarchy:** Implemented DMA-based loading and dual-port SRAM buffering to support up to 16x16 kernel execution.
  - **Physical Design:** Executed RTL-to-GDSII synthesis using OpenLane2, validating physical feasibility and achieving timing closure.
- **AES Encryption/Decryption Engine** | *Verilog, FPGA, Cryptography* 2023
  - **Core Logic:** Synthesized robust AES cryptographic core in Verilog HDL supporting 128-bit, 192-bit, and 256-bit key standards.
  - **Datapath:** Engineered hardware-optimized modules for SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
  - **Validation:** Verified cryptographic correctness against standard NIST test vectors using rigorous debug tracing.
- **Tactic (Robotics & AI)** | *Embedded Systems* 2024
  - **System Integration:** Developed browser-controlled robotics with AI vision and low-latency control
  - **Firmware:** Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control.
  - **Backend Processing:** Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.
- **Hankers (Social Platform)** | *Full-Stack Engineering* 2025
  - **Architecture:** Built scalable social platform using Microservices principles, React.js, and TypeScript.
  - **Frontend:** Implemented responsive UI design and asynchronous API integration via AJAX.
  - **DevOps:** Established CI/CD pipelines and containerized services using Docker for consistent deployment.
- **x86 Brick Breakers Arcade** | *Assembly (x86), MASM, Low-Level Programming* 2024
  - **Game Logic:** Developed a real-time, multiplayer arcade game using x86 Assembly and the MASM assembler.
  - **System Control:** Directly manipulated video memory for low-latency graphics rendering and managed hardware interrupts to handle real-time I/O.
- **Alien Invasion** | *Combat Simulation, C++* 2023
  - **Simulation:** Modeled combat scenarios using custom data structures to simulate fleet mechanics and collision detection.

## STUDENT ACTIVITIES

- **IEEE** | *UI/UX Designer - Frontend Team Leader* Cairo | Oct 2025 – Present
  - **Portfolio Platform Development:** Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- **NASA Space Apps Challenge** | *Participant – Project: "Orrery"* Cairo | Oct 2024
  - **Orrery App:** Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

## TECHNICAL SKILLS

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- **Digital Design & Verification:** Verilog HDL, VHDL, RTL Design, RISC-V ISA, MIPS Architecture, Pipeline Design, RTL-to-GDSII Flow, OpenLane2, DRC/LVS Verification, Static Timing Analysis, FSM Design
- **Development Tools & Platforms:** ModelSim, QuestaSim, Intel Quartus Prime, GTKWave, ESP32, AVR/Arduino
- **Programming Languages:** C/C++ (Advanced), Python, x86 Assembly, MIPS Assembly